EViews — Econometric Analysis of Panel Data Training Course

Introduction.
This course offered at IRES is designed to equip participants with the tools and techniques needed to conduct econometric analysis of panel data using EViews. Panel data, which combines time series and cross-sectional dimensions, is a powerful tool for analyzing the behaviour of entities over time. The course covers a range of topics, including fixed and random effects models, dynamic panel data models, and testing for heterogeneity and autocorrelation. Participants will learn how to use EViews to manage and analyse panel data, interpret model outputs, and apply these techniques to real-world data.

Course Duration: 5 Days

Target Audience:

- Economists
- Financial analysts
- Policy analysts
- Researchers
- Statisticians
- Data scientists and professionals involved in panel data analysis

Personal Impact:

- Develop expertise in econometric modeling of panel data using EViews.
- Enhance your ability to work with complex datasets that involve both time series and cross-sectional components.
- Gain practical experience in analysing and interpreting panel data results.

Organizational Impact:

- Improve the organization's ability to analyze longitudinal data for better decision-making.
- Apply panel data techniques to evaluate performance, trends, and policy impacts.
- Utilize econometric models to improve business forecasting and strategic planning.

Course Level: Foundation

Course Objectives:
By the end of this course, participants will be able to:

- Understand the structure and key principles of panel data econometrics.
- Estimate and interpret fixed and random effects models using EViews.
- Build and estimate dynamic panel models, including system GMM.
- Test for heterogeneity, autocorrelation, and other issues in panel data.
- Apply panel data econometrics to real-world economic, financial, and policy-related datasets.

Course Outline:

Module 1: Introduction to Panel Data and EViews

- Overview of panel data and its structure
- Introduction to EViews for panel data analysis
- Importing, managing, and organizing panel data in EViews
- Practical Exercise: Data management for panel data in EViews

Module 2: Fixed Effects and Random Effects Models

- Understanding fixed and random effects models
- Estimating and interpreting fixed effects models in EViews
- Estimating random effects models and running the Hausman test
- Case Study: Application of fixed and random effects models using firm-level data

Module 3: Dynamic Panel Data Models

- Introduction to dynamic panel data models
- Using difference and system GMM for dynamic modeling
- Dealing with endogeneity and autocorrelation in dynamic models
- Practical Exercise: Estimating dynamic panel models using EViews

Module 4: Diagnostic Tests for Panel Data Models

- Testing for heterogeneity, autocorrelation, and heteroscedasticity in panel data
- Conducting diagnostic tests in EViews

- Interpreting model outputs and refining models
- Case Study: Conducting diagnostic testing and model refinement for economic data

Module 5: Advanced Applications of Panel Data Analysis

- Advanced topics in panel data econometrics, such as cointegration and causality tests
- Practical Exercise: Advanced panel data modeling in EViews
- Real-life Project: Developing and interpreting a panel data model using your organization's data